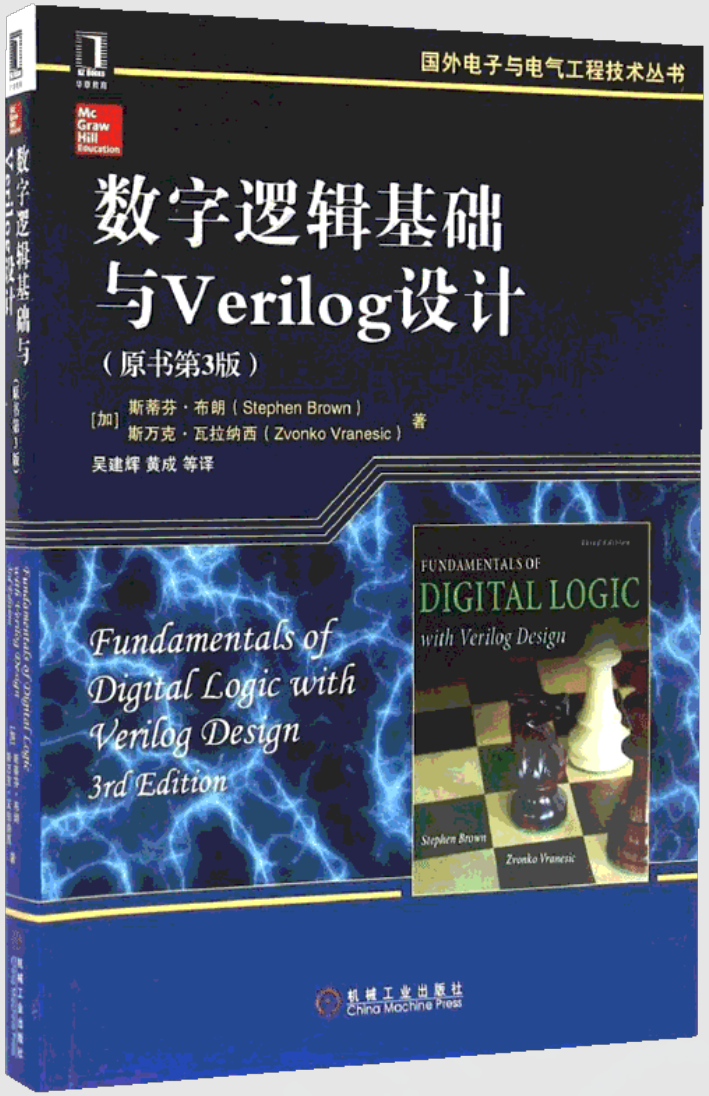
深圳理工大学 SHENZHEN UNIVERSITY OF ADVANCED TECHNOLOGY



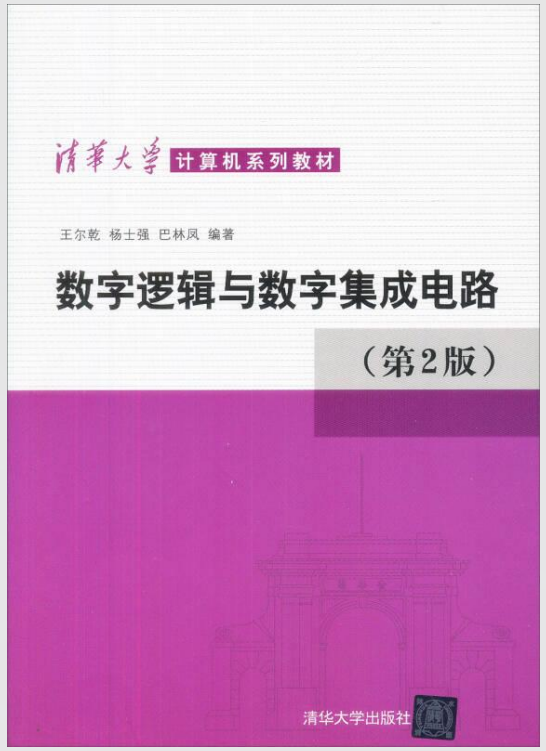
Course Introduction

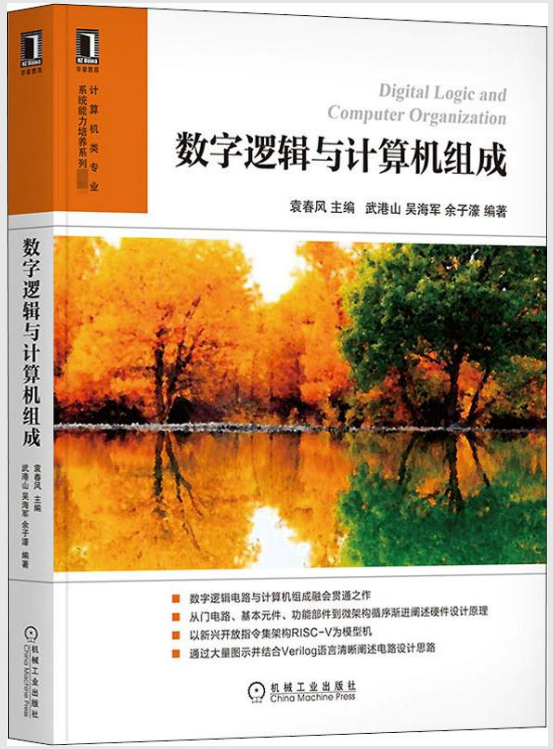
陈奕含
Faculty of Computility Microelectronics

Textbook



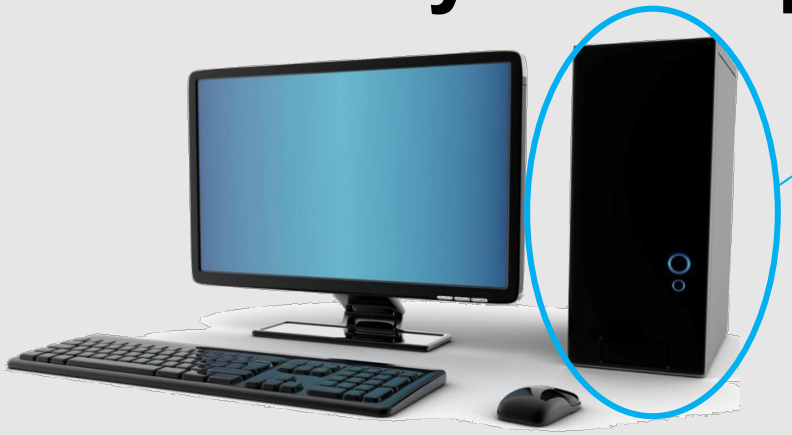
Ref book

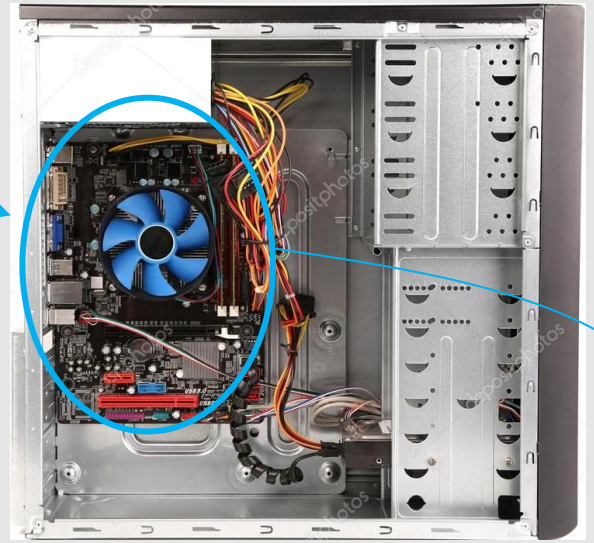




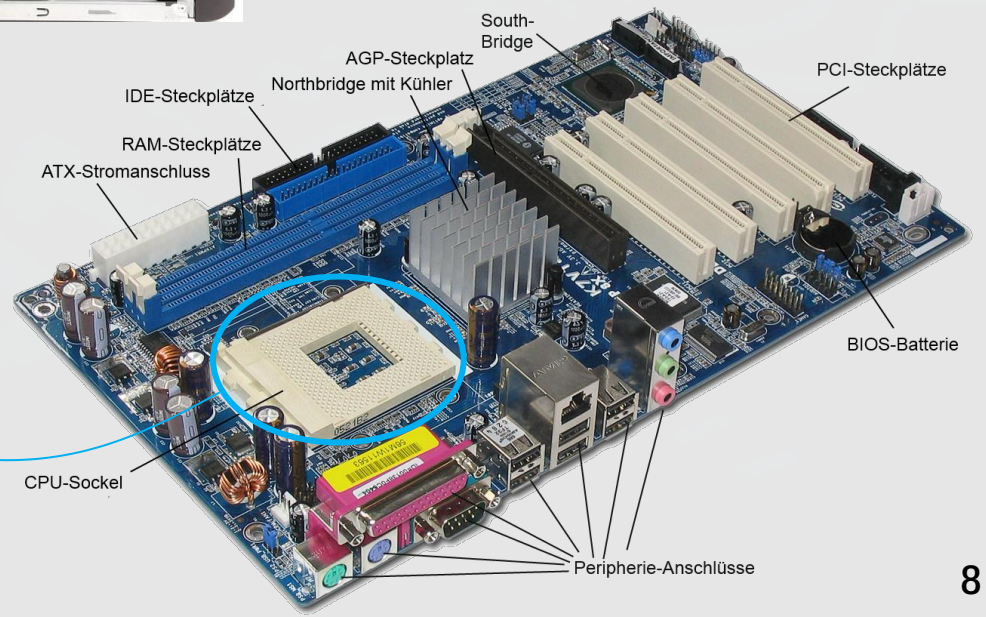
What's the Course about?

- **Hierarchy of computer**





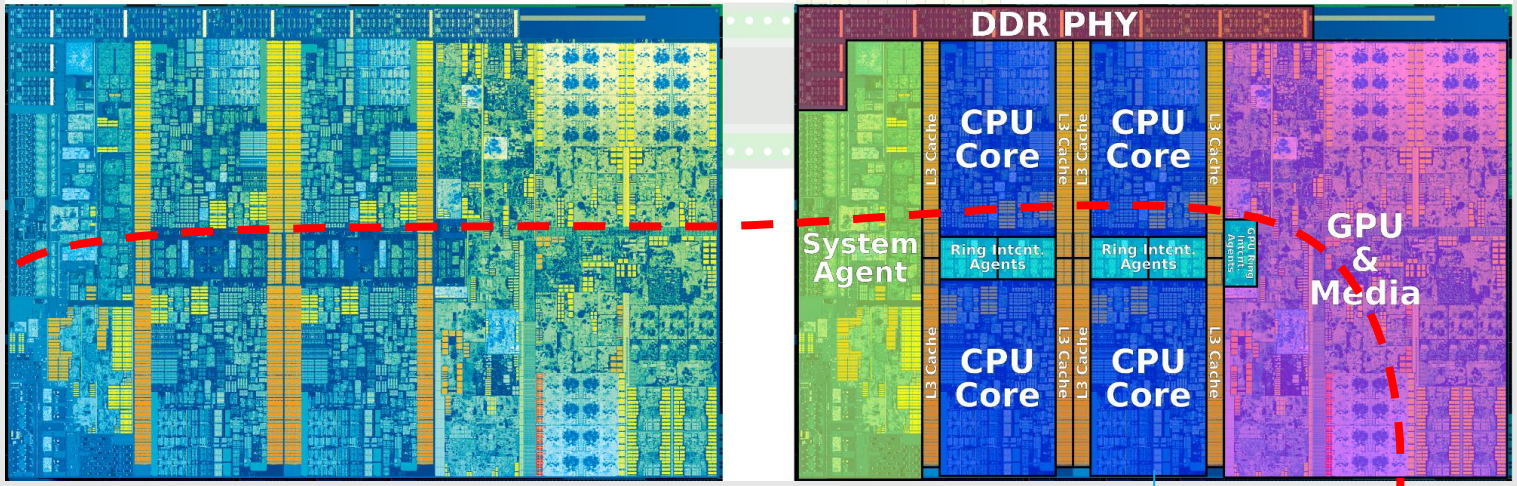


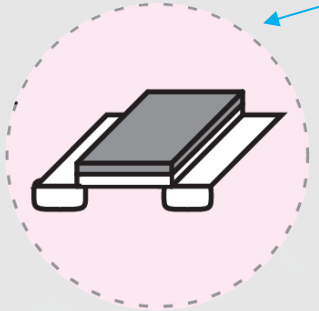


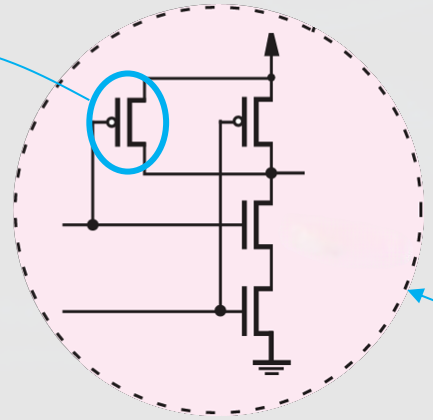
What's the Course about?

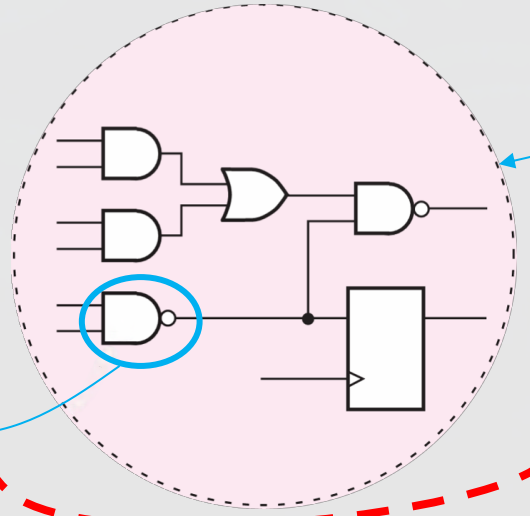
- **Hierarchy of computer**





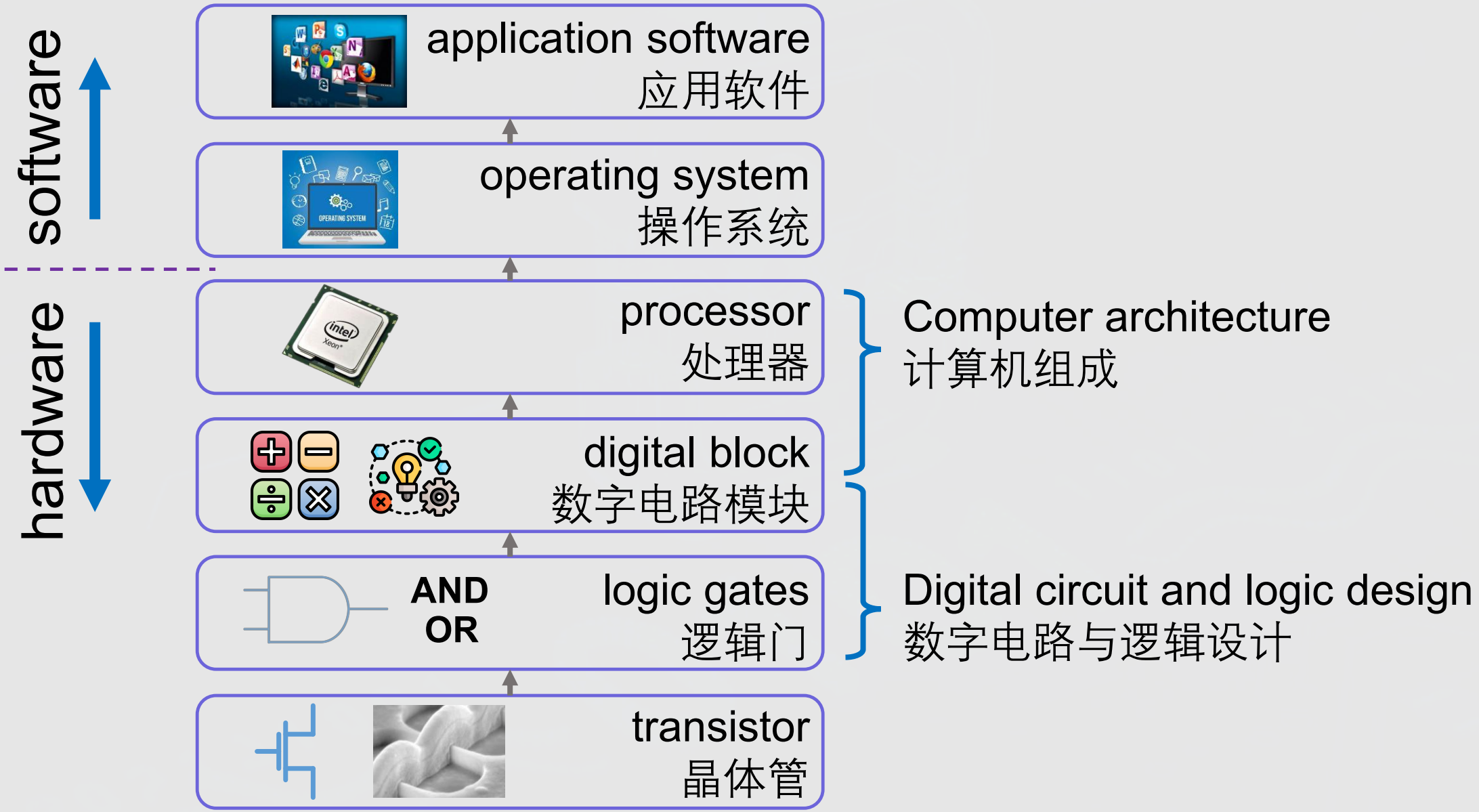






focus of this course

What's the Course about?



Digital

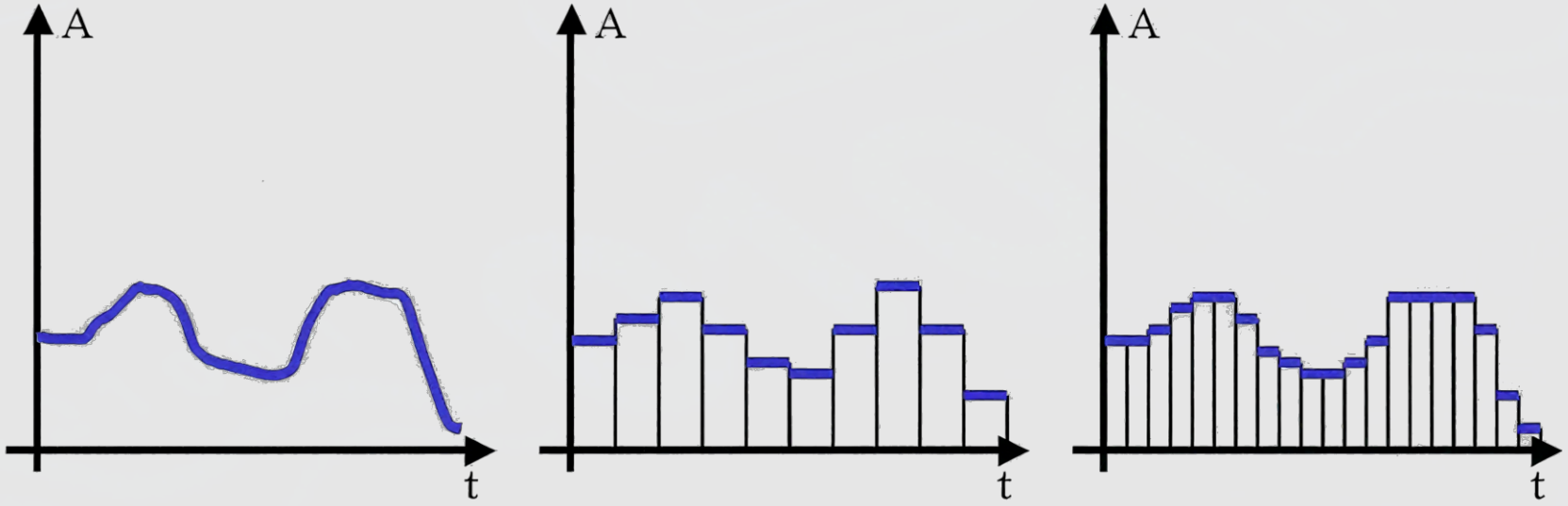
- **Digital vs. Analog 数字与模拟**

Analog signal: continuous value



Digital signal: discrete value

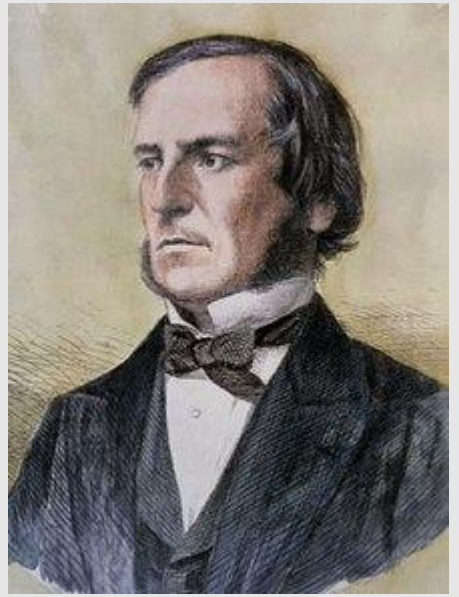


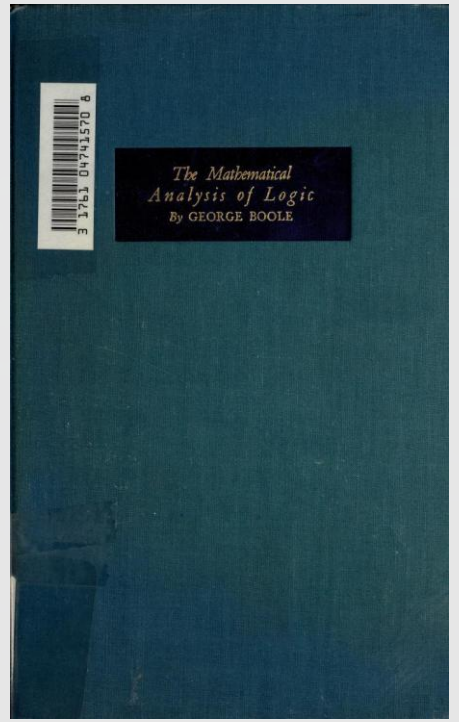


Logic

- **Boolean Algebra (布尔代数)**
  - By British mathematician **George Boole** in his first book “ The Mathematical Analysis of Logic ” (1847)
  - Using mathematical language (“and,” “or,” “not”) to describe the fundamental laws of logical reasoning
  - Purely an achievement in the fields of philosophy and mathematics

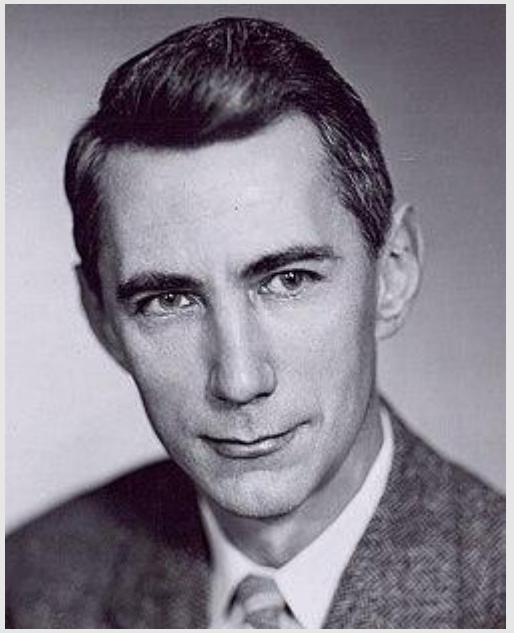
    单纯哲学和数学领域的成果

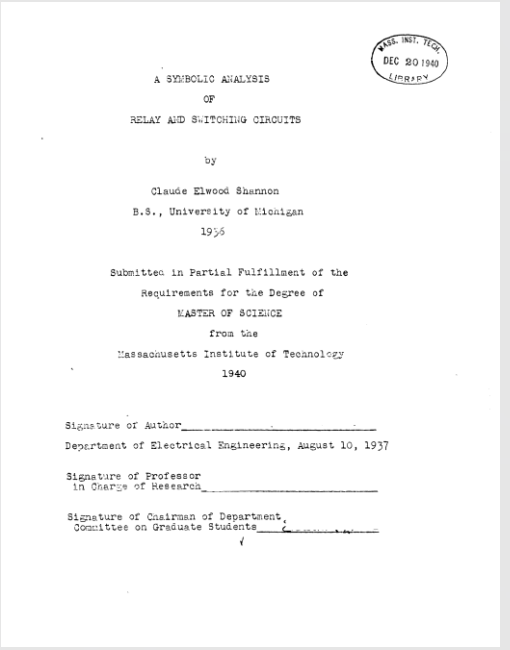




Logic

- **Boolean Algebra for Circuit Systems**
  - By “father of information theory” 信息理论之父 **Claude Elwood Shannon** in his master's thesis “A Symbolic Analysis of Relay and Switching Circuits” (1937)
  - Find correspondence between Boolean algebra's “true/false” and circuit systems' “on/off ” states
  - A milestone in integrating abstract mathematical theory with practical engineering applications





MASS. INST. TECH. DEC 20 1940 LIBRARY

A SYMBOLIC ANALYSIS
OF
RELAY AND SWITCHING CIRCUITS

by

Claude Elwood Shannon
B.S., University of Michigan
1936

Submitted in Partial Fulfillment of the
Requirements for the Degree of
MASTER OF SCIENCE
from the
Massachusetts Institute of Technology
1940

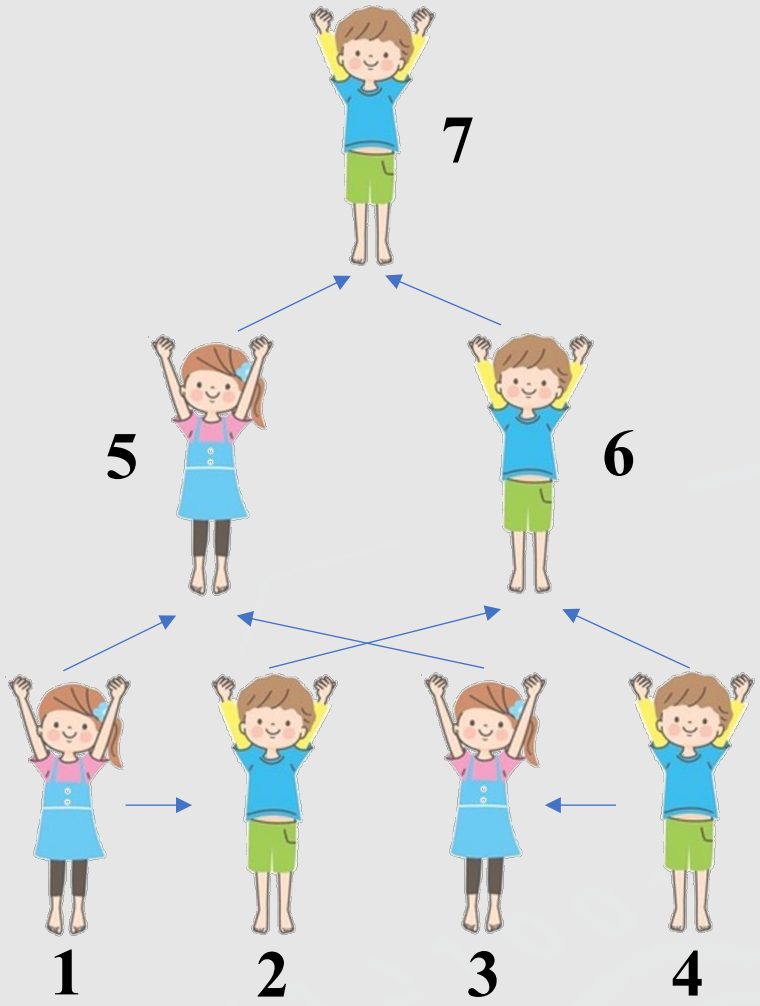
Signature of Author_____
Department of Electrical Engineering, August 10, 1937

Signature of Professor
in Charge of Research_____

Signature of Chairman of Department
Committee on Graduate Students_____

What Can Logic Do ?

- **Activity Demo**



Instruction for kid No.1 :

- if the teacher says “1”, you crouch　老师说“1”，蹲
- if the teacher says “2”, you crouch　老师说“2”，蹲
- if the teacher says “3”, you stand　老师说“3”，站
- if the teacher says “4”, you stand　老师说“4”，站

Instruction for kid No.4 :

- if the teacher says “1”, you crouch　老师说“1”，蹲
- if the teacher says “2”, you stand　老师说“2”，站
- if the teacher says “3”, you crouch　老师说“3”，蹲
- if the teacher says “4”, you stand　老师说“4”，站

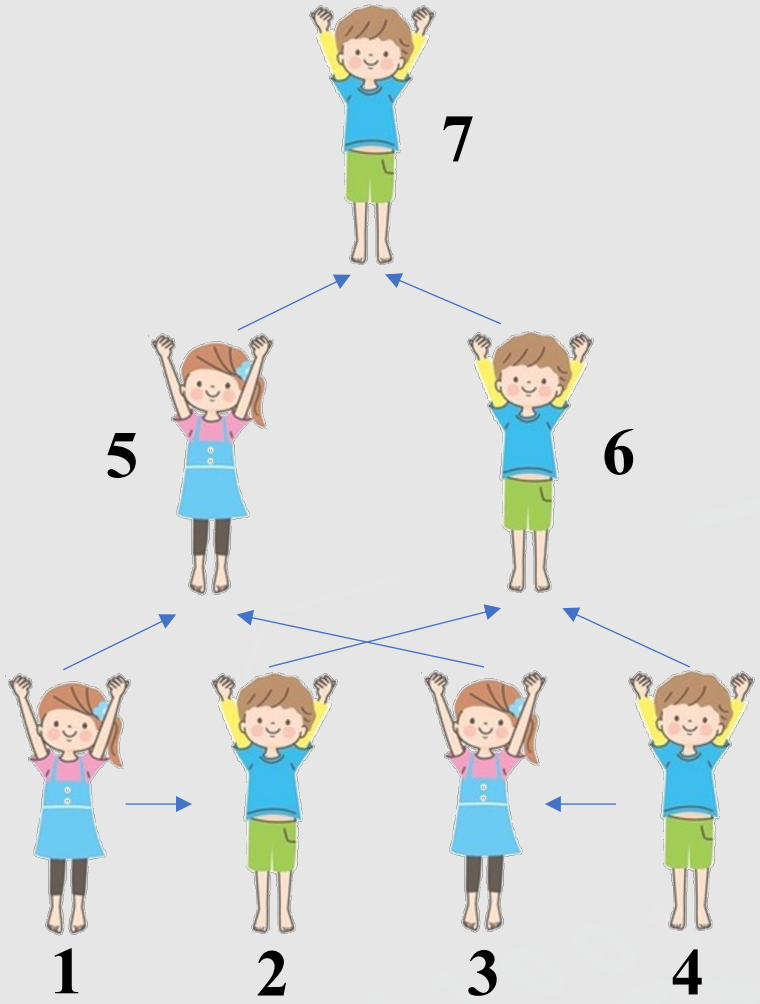
Instruction for kid No.2 :

- if No.1 crouch, you stand　1号蹲，你就站
- if No.1 stand, you crouch　1号站，你就蹲

Instruction for kid No.3 :

- if No.4 crouch, you stand　4号蹲，你就站
- if No.4 stand, you crouch　4号站，你就蹲

What Can Logic Do ? - Activity Demo



Instruction for kid No.5 :

- ❖ if anyone of No.1 and No. 3 is crouching, you crouch
- ❖ if they are both standing, you stand
- ❖ 1号和3号只要有蹲着的，你就蹲
- ❖ 当两个人都站着，你就站

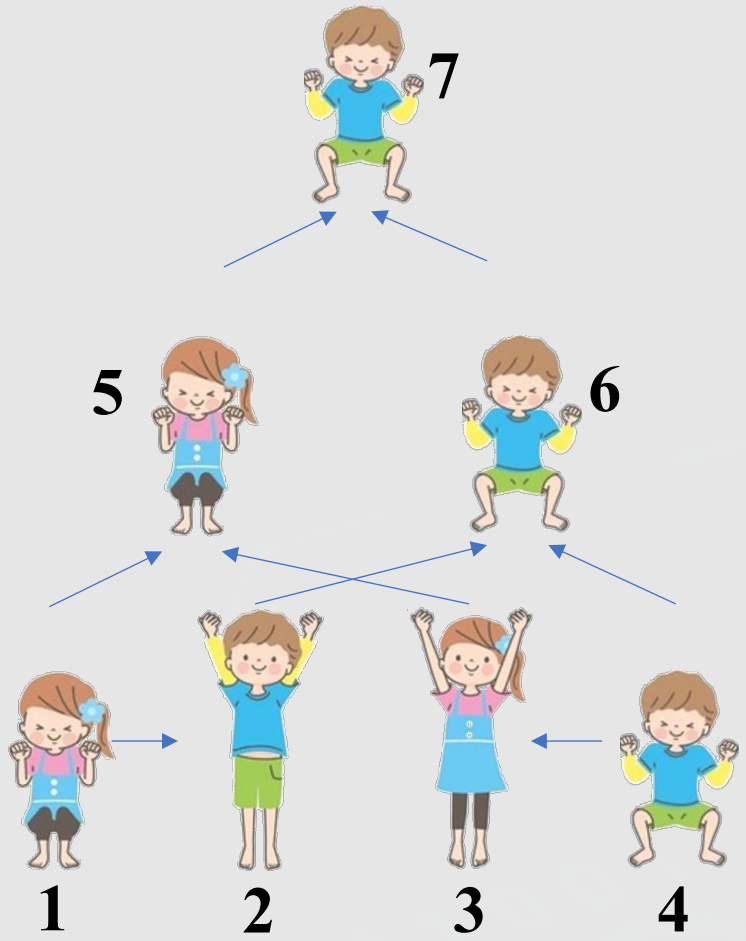
Instruction for kid No.6 :

- ❖ if anyone of No.2 and No. 4 is crouching, you crouch
- ❖ if they are both standing, you stand
- ❖ 2号和4号只要有蹲着的，你就蹲
- ❖ 当两个人都站着，你就站

Instruction for kid No.7 :

- ❖ if anyone of No.5 and No. 6 is standing, you stand
- ❖ if they are both crouching, you crouch
- ❖ 5号和6号只要有着站的，你就站
- ❖ 当两个人都蹲着，你就蹲

What Can Logic Do ? - Activity Demo

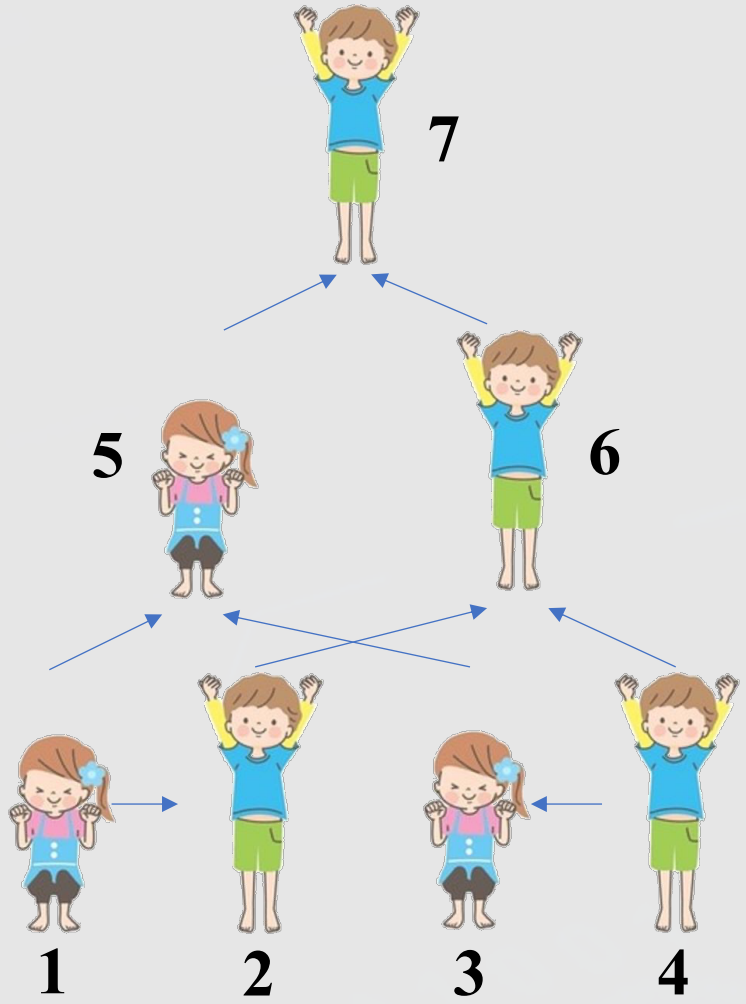


Record the gestures of No.1, No.4 and No. 7

- Mark “1” for standing
- Mark “0” for crouching

Step	No. 1	No. 4	No. 7
1	0	0	0

What Can Logic Do ? - Activity Demo

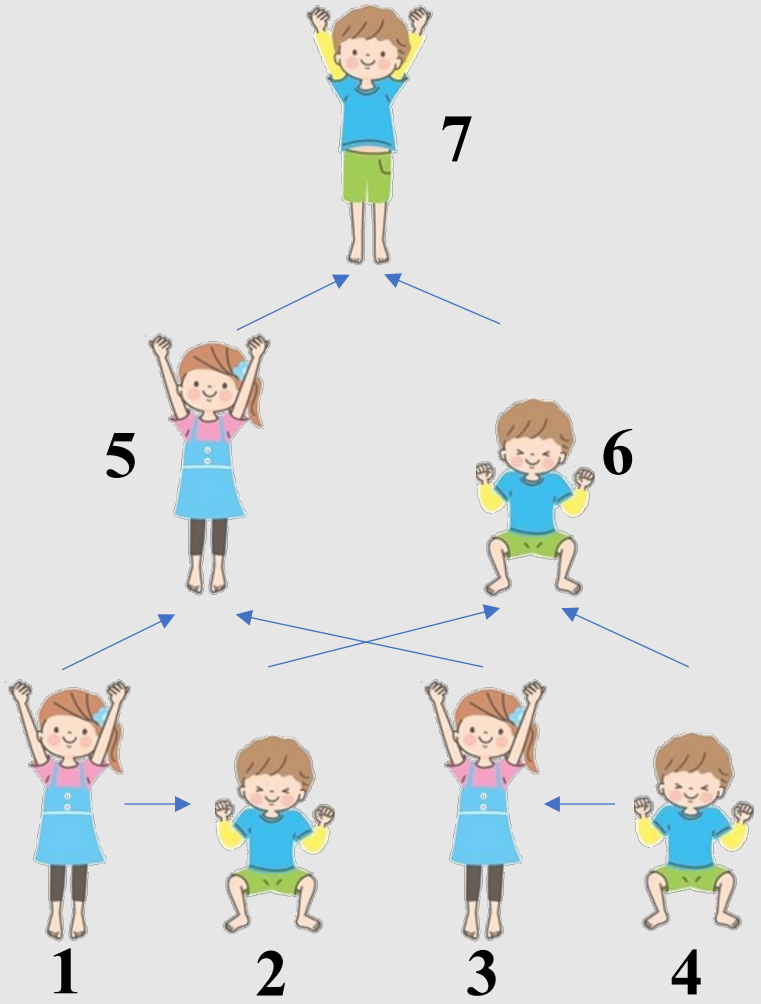


Record the gestures of No.1, No.4 and No. 7

- Mark “1” for standing
- Mark “0” for crouching

Step	No. 1	No. 4	No. 7
1	0	0	0
2	0	1	1

What Can Logic Do ? - Activity Demo

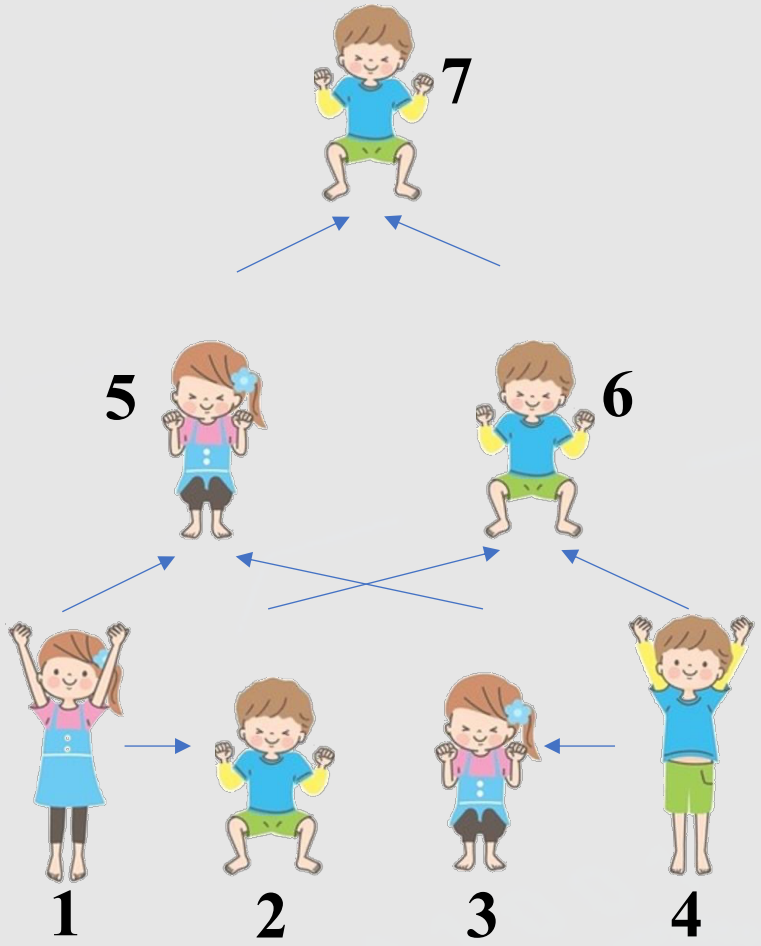


Record the gestures of No.1, No.4 and No. 7

- Mark “1” for standing
- Mark “0” for crouching

Step	No. 1	No. 4	No. 7
1	0	0	0
2	0	1	1
3	1	0	1

What Can Logic Do ? - Activity Demo



Record the gestures of No.1, No.4 and No. 7

- Mark “1” for standing
- Mark “0” for crouching

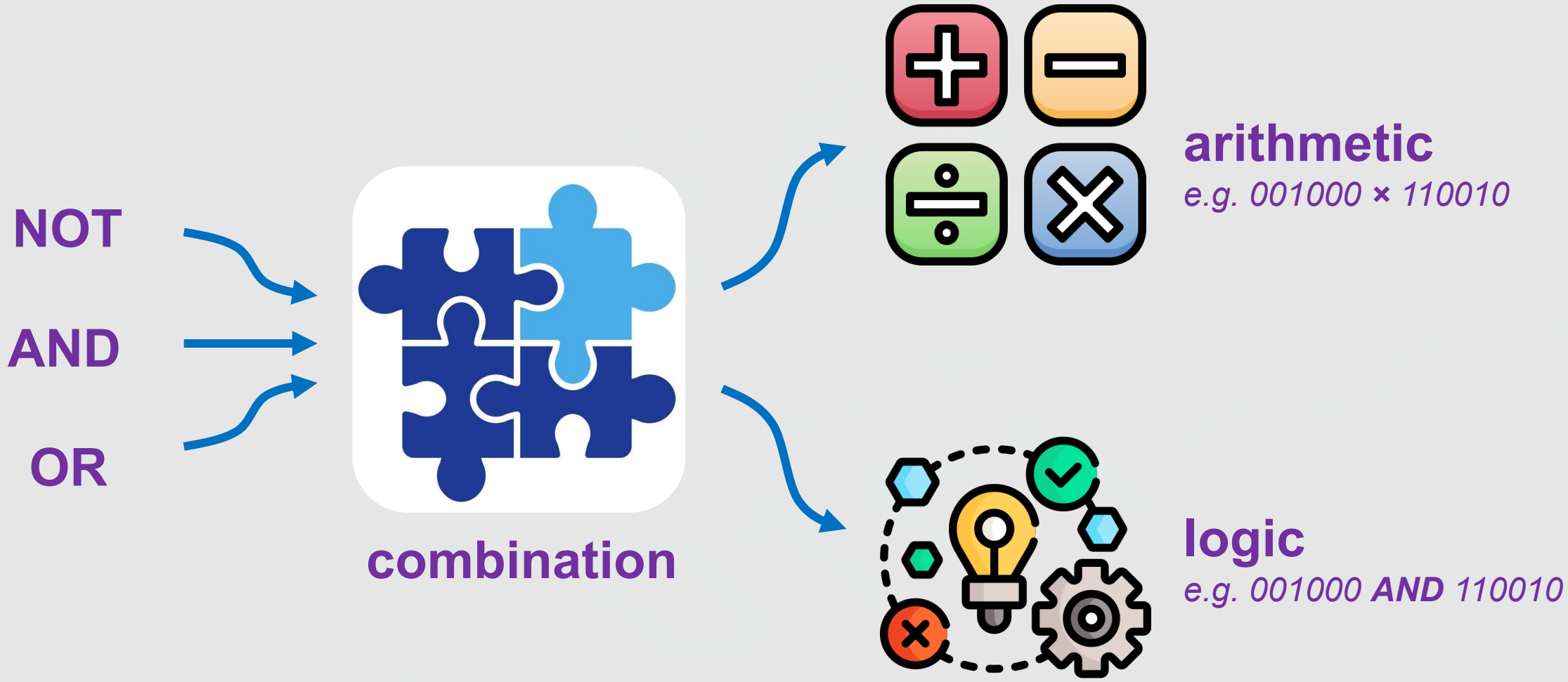
Step	No. 1	No. 4	No. 7
1	0	0	0
2	0	1	1
3	1	0	1
4	1	1	0

What Can Logic Do ? - Activity Demo

Step	No. 1	No. 4	No. 7	No.1 + No.4
1	0	0	0	0
2	0	1	1	1
3	1	0	1	1
4	1	1	0	1 0

What Can Logic Do ?

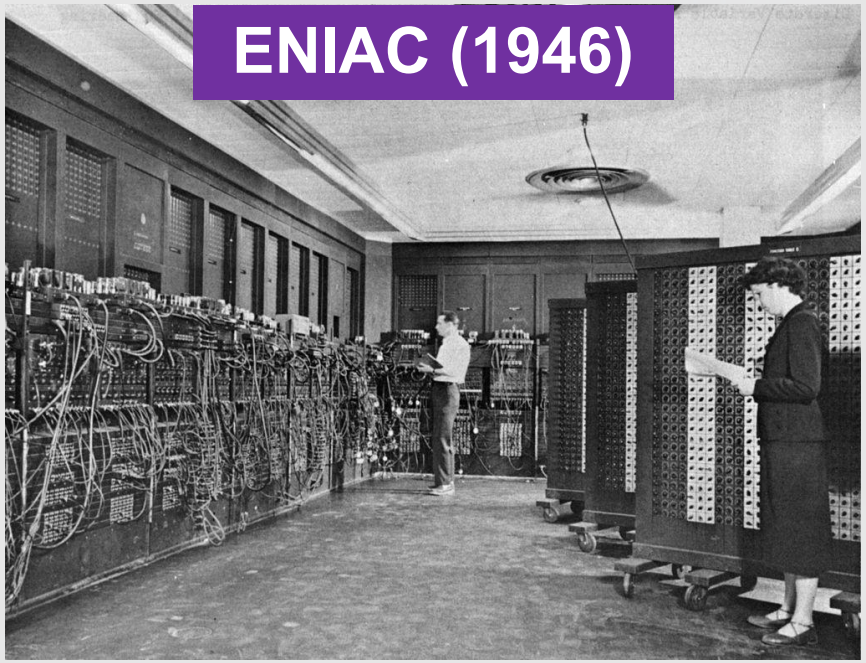
- **Logic for Arithmetic and Logic Control**

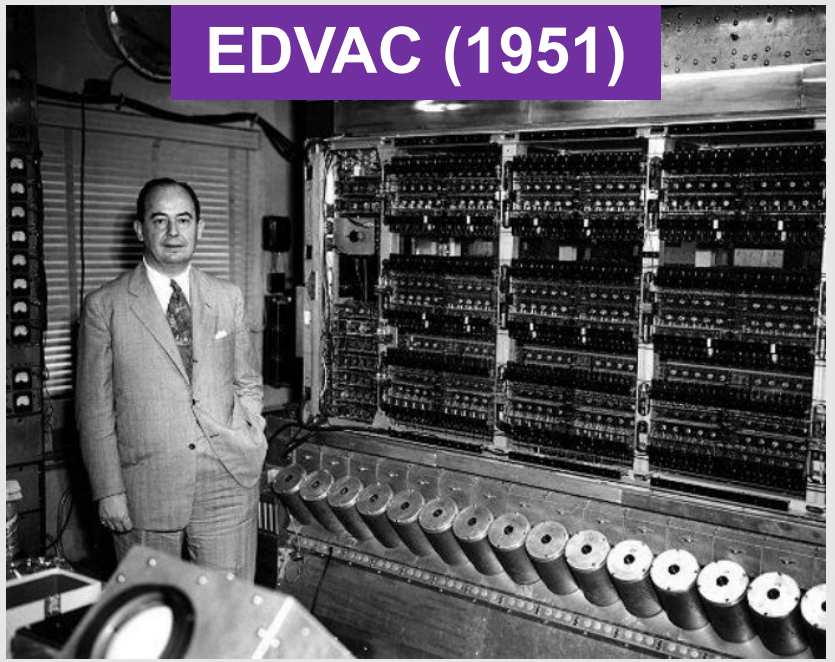


What for Logic ?



- **Early Implementation of Logic**
  - By relays (继电器) and later vacuum tubes (真空管)
  - ENIAC (the world's first general-purpose electronic computer) utilized ≈18,000 vacuum tubes

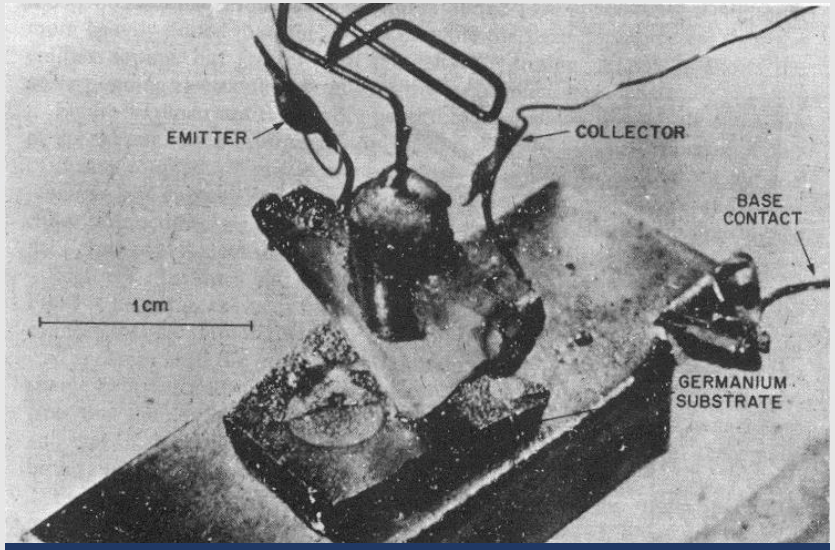




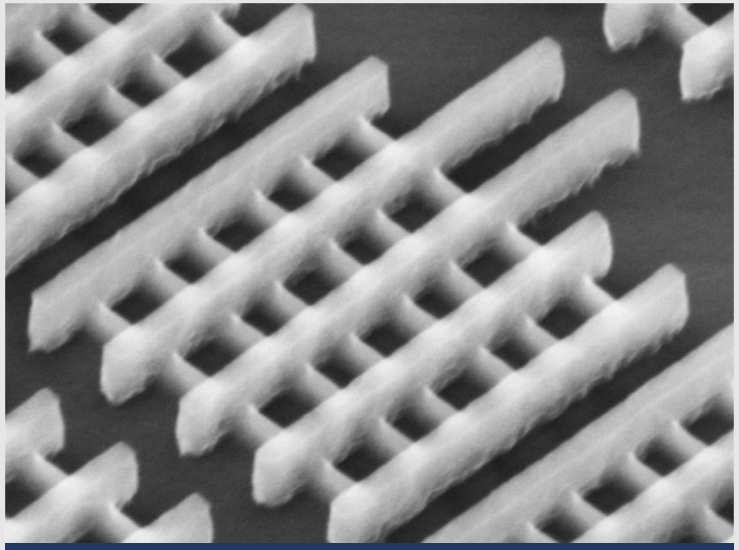
What for Logic ?

- **Transistor (晶体管) for Logic**
  - Co-invented by John Bardeen, William Shockley, Walter Brattain (1947)
  - 1956 Nobel Prize in Physics





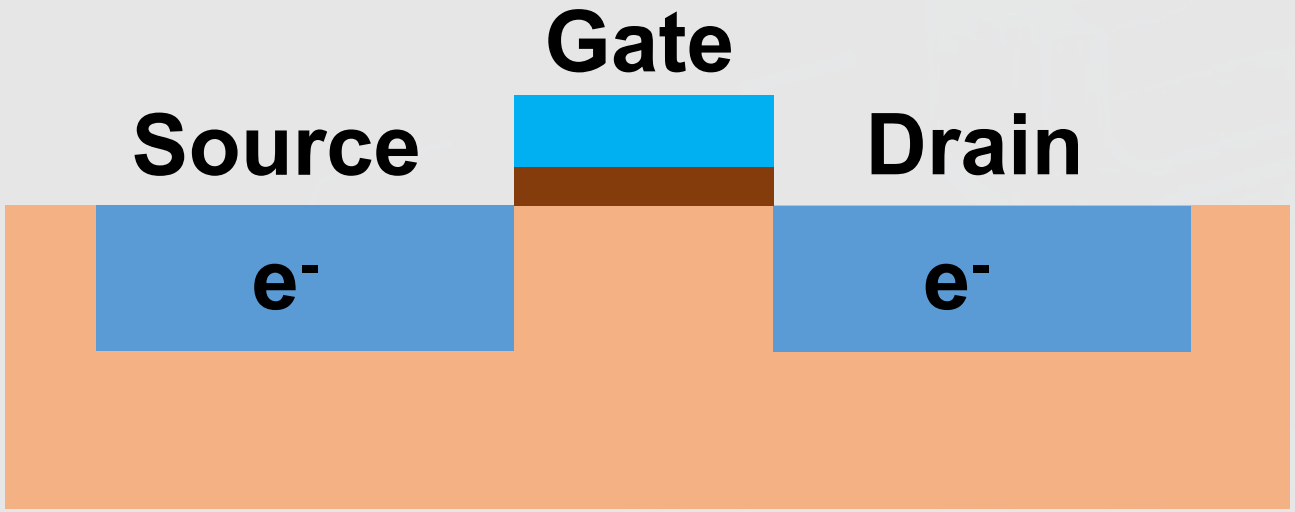
1947 First Transistor

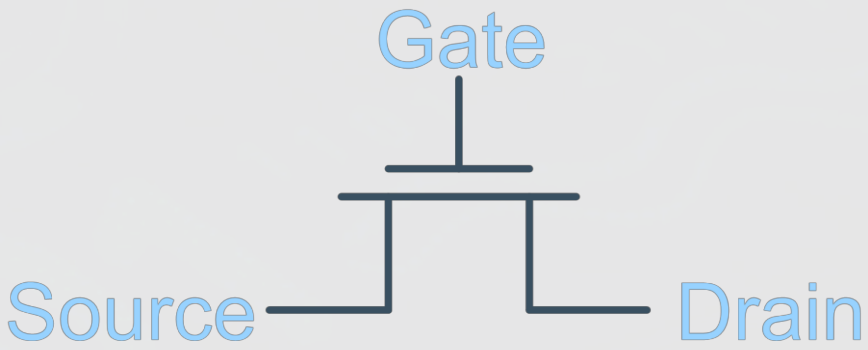


22nm Tri-gate Transistor

What for Logic ?

- **MOSFET Transistor:**
  - A semiconductor material as substrate
  - Create electron ocean: source & drain
  - Create controlling gate

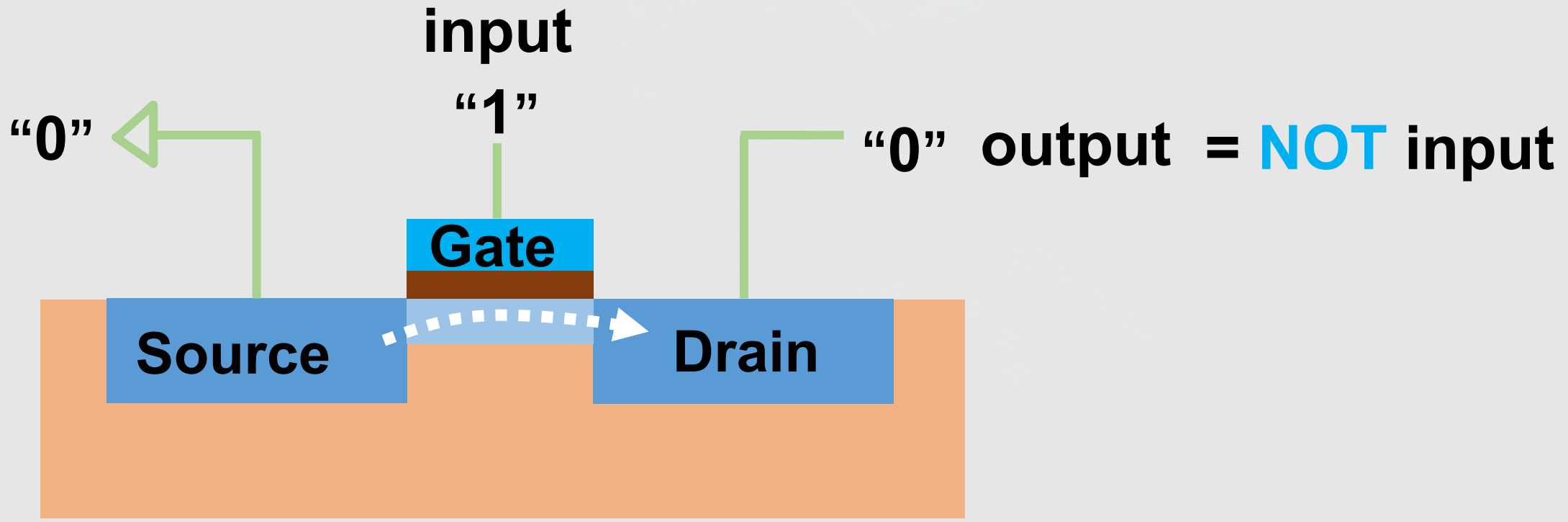




What for Logic ?

- **MOSFET Transistor:**

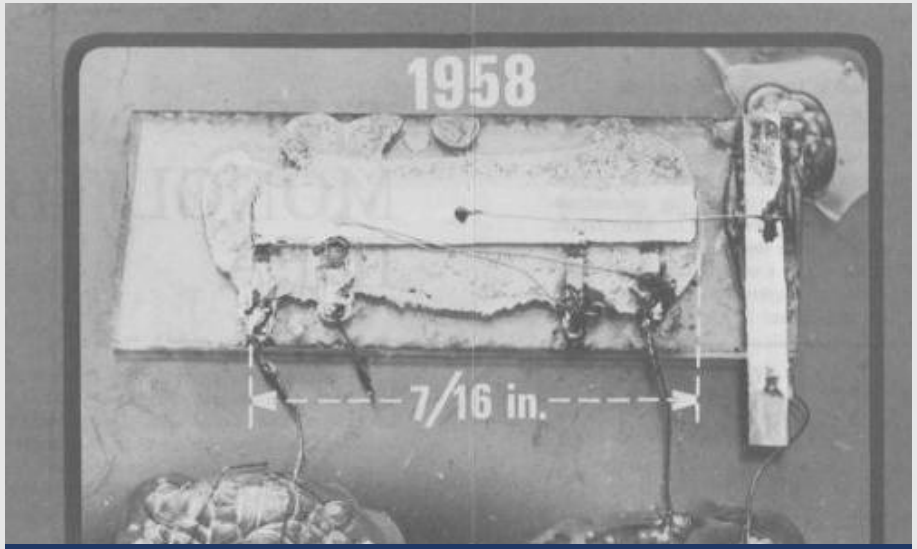
  Gate = high voltage “1”, channel is formed for e^-
  With grounded source, drain = low voltage “0”



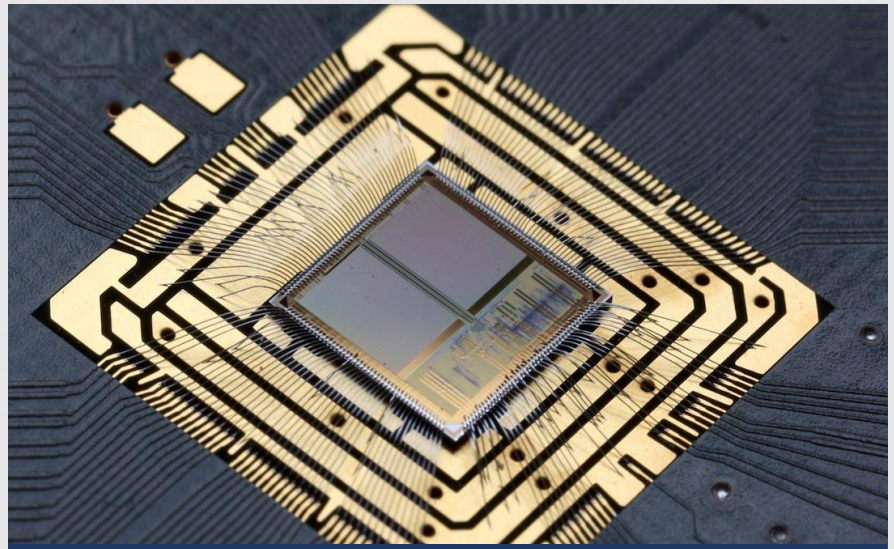
Logic Circuits

- **Integrated Circuit (IC 集成电路)**
  - July 1958, **Jack Kilby**, Texas Instruments,
  - Half a year after, **Robert Noyce**, Fairchild Semiconductor
  - Kilby won the 2000 Nobel Prize in Physics





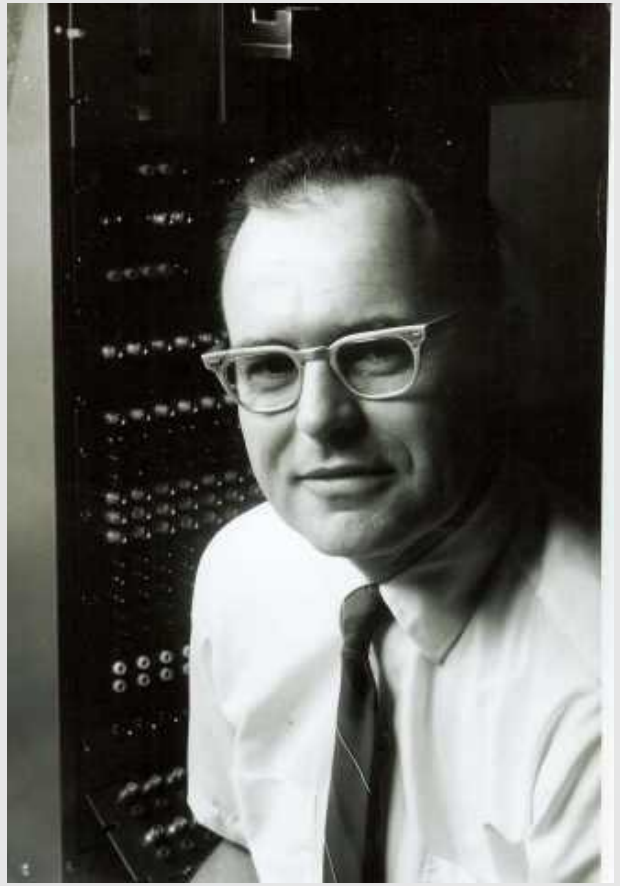
1958 First IC

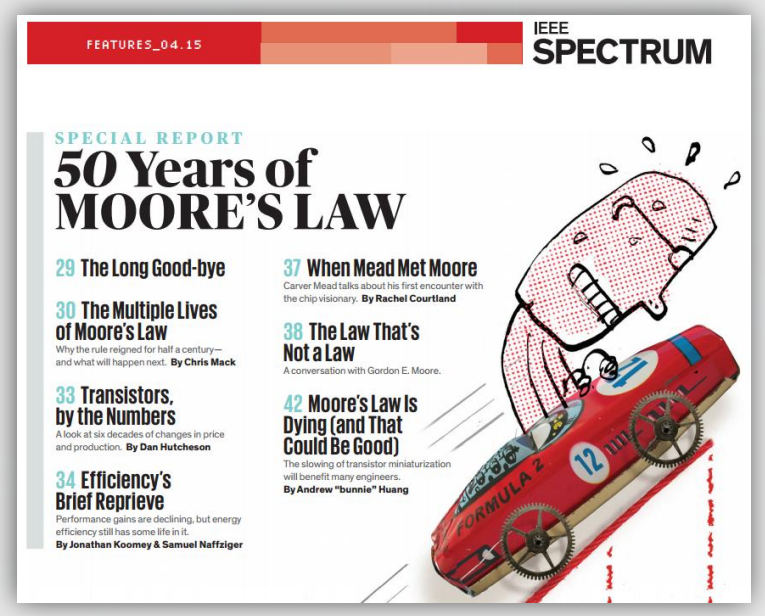


Recent IC

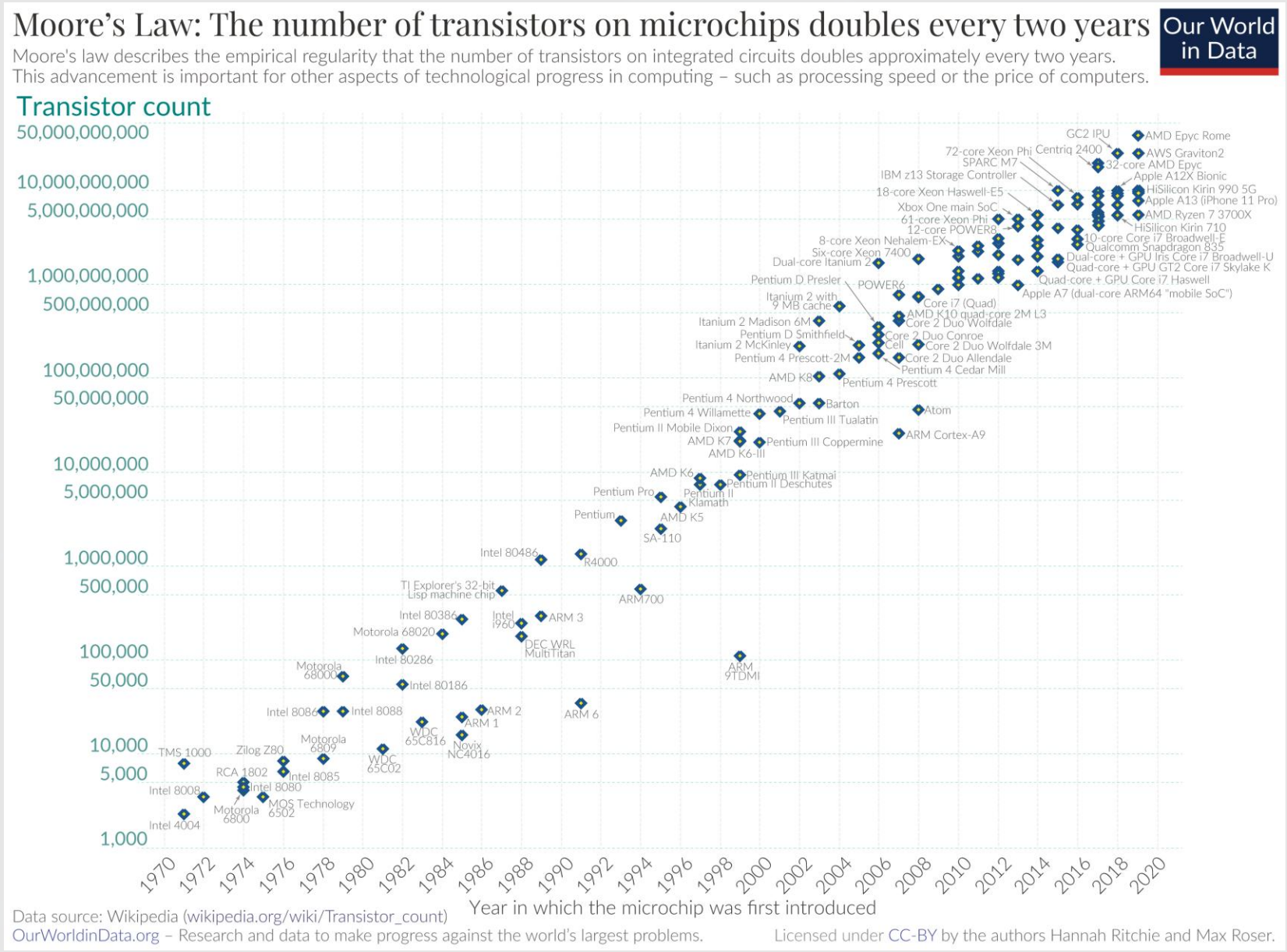
Logic Circuits

- **Moore's Law (摩尔定律)**
  - By **Gordon Moore** (co-founder of Fairchild Semiconductor and Intel 仙童与英特尔的创始人)
  - The **observation** in 1965 that the number of transistors in a dense integrated circuit doubles approximately every two years 每两年翻一番
  - Not a physical or natural law
  - Still work ?





Logic Circuits



Advances of Logic Circuits

- **More Moore:**
  - New MOSFET structure: FinFET and GAAFET
- **New architecture:**
  - Compute-in-memory
  - Neuromorphic computing
- **New material**
  - Beyond Silicon: graphene, MoS_2
  - Spintronics
- **Integration and Packaging**
  - SoC
  - 3D integration

End of Introduction

Formal journal begins!